

Validation of Calibration



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1 Objective

To Validate the calibration of radiation survey instruments and Pocket Dosimeters.

2 Apparatus

- Radiation Survey Meter
- Digital Contamination Monitor
- Micro Digital Pocket Dosimeter
- Pressurized micro-R ion chamber survey meter
- Cs-137 source with Lead Shield
- Calibration Bench and other accessories

3 Theory

Radiation Survey meters and Contamination Monitors are hand-held ionizing radiation measuring instruments used to check the radiation level in an environment containing radioactive sources or radiation-generating equipment. These instruments use gas-filled radiation detectors, semiconductor radiation detectors, or scintillation detectors. Out of these three, Gas-filled radiation detectors are very commonly used. Gas-filled detectors have three types: the Ionization chamber, the Proportional counter, and the Geiger - Muller (G.M.) counter. They all work in their respective voltage region.

3.1 G.M Based Instruments (General)

3.1.1 Working Principle

Geiger-Müller (G.M) based instruments operate on the principle of gas ionization in the *Geiger region*.

- **Townsend Avalanche:** The detector consists of a gas-filled tube (usually Argon or Neon with a Halogen quenching agent) with a central anode wire maintained at a high positive potential relative to the cathode wall.
- **Ionization:** When ionizing radiation enters the tube, it creates ion pairs (electrons and positive ions). The strong electric field accelerates electrons toward the anode, causing secondary ionizations.
- **Geiger Discharge:** This results in a massive "avalanche" of electrons (Townsend avalanche) that spreads along the entire length of the anode wire. This produces a large, detectable electrical pulse independent of the energy of the incident radiation.
- **Quenching:** To prevent continuous discharge, a quenching gas (e.g., Chlorine or Bromine) absorbs the energy of positive ions reaching the cathode, stopping the emission of secondary electrons.

3.1.2 Application in Medical Physics

- **Leakage and Spill Detection:** Used for rapid detection of radioactive spills or leaks in nuclear medicine hot labs.
- **Locating Lost Sources:** Due to their high sensitivity (but low efficiency for gamma), they are ideal for locating lost low-activity seeds (e.g., I-125 seeds in Brachytherapy).
- **General Survey:** Used for "go/no-go" checks to verify if an area is radiation-free.



Fig.1. Radiation Survey Meter



Fig.2 Thick-walled G. M. Tube Detector

3.2 Radiation Survey Meter [Model: RM701N]

3.2.1 Working Principle

The Model RM701N is a microcontroller-based survey meter that utilizes an **Energy Compensated G.M. Tube** (typically Model GM-130 or equivalent).

- **Energy Compensation:** Unlike standard G.M tubes which over-respond to low-energy gamma rays, this detector is wrapped in a filter (usually tin or lead) to flatten the energy response. This ensures the count rate is proportional to the dose rate across a wide energy range (e.g., 60 keV to 1.25 MeV).
- **Processing:** The instrument counts the pulses from the detector and uses a calibration factor to convert the count rate (CPM) into dose rate units such as mR/hr or $\mu\text{Sv/hr}$.

3.2.2 Application in Medical Physics

- **Area Monitoring:** Routine surveys of corridors and waiting rooms adjacent to radiation therapy bunkers or X-ray suites.
- **Package Inspection:** Checking transport packages (receiving radioactive isotopes) to ensure surface dose rates comply with transport regulations (e.g., Transport Index).
- **Waste Disposal:** Verifying that waste from nuclear medicine patients has decayed to background levels before disposal.

3.3 Digital Contamination Monitor [Model: CM710N]

3.3.1 Working Principle

The CM710N is designed specifically for detecting surface contamination and typically uses a **Pancake G.M. Probe** (e.g., GM-138).

- **Detector Geometry:** The "pancake" shape provides a large surface area for detection.
- **Thin Window:** It features a very thin mica window ($1.5 - 2.0 \text{ mg/cm}^2$), which allows alpha particles, low-energy beta particles, and gamma rays to enter the detector and cause ionization.
- **Output:** The readout is typically in Counts Per Minute (CPM) or Counts Per Second (CPS), which is directly related to the surface activity (Becquerels) if the efficiency is known.

3.3.2 Application in Medical Physics

- **Personnel Monitoring:** Checking hands, feet, and clothing of staff leaving the "active" area in Nuclear Medicine.



Fig.3 Digital Contamination Monitor



Fig.4 Pancake GM detector

- **Surface Contamination Checks:** Monitoring workbenches, fume hoods, and floor areas for radioactive contamination (e.g., Tc-99m, I-131).
- **Decontamination Verification:** Confirming that a spill has been successfully cleaned up.

3.4 Micro-digital Pocket Dosimeter [Model: MPD-1501]

3.4.1 Working Principle

The MPD-1501 is an Electronic Personal Dosimeter (EPD) that typically uses a **Silicon Diode (Semiconductor)** or a miniature energy-compensated G.M. tube.

- **Semiconductor Mode:** Radiation creates electron-hole pairs in the depletion region of the silicon diode. These charges are collected, amplified, and counted.
- **Real-Time Integration:** A microprocessor integrates the pulses over time to calculate the accumulated dose (μSv or mrem) and the current dose rate.
- **Alarm Thresholds:** The device has programmable thresholds for dose and dose rate, triggering an audible alarm if limits are exceeded.

3.4.2 Application in Medical Physics

- **Real-Time Staff Dose Monitoring:** Worn by staff during high-dose procedures (e.g., Interventional Radiology, Brachytherapy loading) to monitor exposure in real-time.
- **ALARA Enforcement:** Helps keeping doses "As Low As Reasonably Achievable" by warning staff immediately when they enter a high-radiation field.
- **Pregnancy Monitoring:** Essential for monitoring the fetal dose of pregnant staff members to ensuring it remains below regulatory limits.

3.5 451P Pressurized μR Ion Chamber Survey Meter

3.5.1 Working Principle

The 451P is an ionization chamber instrument operating in the **Ion Saturation Region**.

- **Pressurized Gas:** The chamber is filled with Argon gas pressurized to approximately 8 atmospheres (125 psi).
- **Increased Sensitivity:** The high pressure increases the number of gas molecules, thereby increasing the probability of interaction with ionizing radiation. This allows the measurement of very low exposure rates ($\mu\text{R/hr}$).

- **Direct Collection:** Unlike G.M tubes, there is no gas multiplication. The ions produced by radiation are collected directly by the electrodes, creating a tiny current (femtoto-picoamperes) proportional to the energy deposited.
- **Flat Energy Response:** It provides an accurate measurement of exposure (Roentgen) effectively independent of radiation energy (above 25 keV).

3.5.2 Application in Medical Physics

- **Scatter Surveys:** Measuring scatter radiation levels around CT scanners, Fluoroscopy units, and Linear Accelerators (Linacs) for shielding calculations.
- **Shielding Integrity:** Verifying the thickness and effectiveness of lead lining in X-ray room walls and doors.
- **Linear Accelerator Leakage:** Measuring head leakage radiation from Linacs, where G.M counters would saturate due to the pulsed nature of the beam.

3.6 Calibration of Radiation Instruments

Calibration is defined as the quantitative determination of a parameter given by a radiation measuring instrument as a function of the value of the quantity the instrument is intended to measure under a controlled set of standard conditions. The primary objective of calibration is to ensure that the instrument is working properly and, hence, can be used for monitoring purposes. Calibration also helps to find the deviation of the measuring quantity from its actual value, and if needed, the calibration factor can be adjusted based on the overall measurement accuracy.

All the radiation survey and measuring instruments should be calibrated periodically to ensure that they are giving correct/valid readings and it is also a regulatory requirement. A convincing method to calibrate these instruments is to place them in a radiation field of a known dose rate (produced by radioactive elements) and compare their response with the dose rate. Different calibration factors are used for β and γ radiations. The known dose rate can be calculated from the exposure rate constant of the given radionuclide (Calculation shown below).

4 Observation

4.1 Calculation of current activity

Given radioactive source is Cs-137 haing activity 3.7 MBq on September 2022. So the current activity can be calculated using the formula:

$$A = A_0 e^{-\lambda t} \quad (1)$$

Where, A_0 is the initial activity, λ is the decay constant and t is the time elapsed. The decay constant can be calculated using the formula:

$$\lambda = \frac{\ln(2)}{T_{1/2}} \quad (2)$$

Where $T_{1/2}$ is the half-life of the isotope. For Cs-137, $T_{1/2} = 30.0$ years. Calculating λ :

$$\lambda = \frac{\ln(2)}{30.0} = 0.0231 \text{ year}^{-1} \quad (3)$$

Calculating the time elapsed from September 2022 to February 2026:

$$t = 3.33 \text{ years} \quad (4)$$

Now, calculating the current activity:

$$A = 3.7 \text{ MBq} \times e^{-0.0231 \times 3.33} = 3.7 \text{ MBq} \times e^{-0.076923} = 3.7 \text{ MBq} \times 0.9258 = 3.42 \text{ MBq} \quad (5)$$

So, the current activity of the Cs-137 source is approximately 3.418 MBq or 0.0924 mCi.

4.2 True dose rate (R)

The true dose rate can be calculated using the formula:

$$R = \frac{A \cdot \Gamma}{d^2} \quad (6)$$

Where, A is the activity of the source, Γ is the specific gamma ray constant for Cs-137, and d is the distance from the source. The specific gamma ray constant for Cs-137 is approximately $3.2267 \text{ R cm}^2 / \text{mCi h}$. Converting the activity from MBq to Ci:

$$A = 3.418 \text{ MBq} = 0.0924 \text{ mCi} \quad (7)$$

Now, calculating the true dose rate at a distance of 50 centimeter:

$$R = \frac{0.0924 \text{ mCi} \times 3.2267 \text{ R cm}^2 / \text{mCi h}}{(50 \text{ cm})^2} = 118.95 \mu\text{R/h} \quad (8)$$

4.3 Response for Fluke 451 P

Place the fluke 451 P at a distance of 50 cm from the source and measure the dose rate. Different orientations of the detector can be used to measure the dose rate and compare it with the true dose rate calculated above.

Cs-137		A = 3.7 MBq		Sept 2022		Distance = 50 cm			
True dose rate (uR/h)		Average background counts(B) = 16.04					Corrected dose rate	Uncertainty	
118.95	1	2	3	4	5	Avg	103.46	13.02227827	
	112	124	117	147	136	119.5			
	126	109	115	122	129				
	124	106	121	115	114				
	119	112	130	117	116				
	113	111	120	119	125				
	111	115	123	121	117				
	117.5	112.8	121.0	123.5	122.8				

True dose rate (uR/h)	Average background counts(B) = 16.04						Corrected dose rate	Uncertainty	
	1	2	3	4	5	Avg			
118.95	122	130	118	125	113	129.3	113.26	4.783522488	
	131	138	131	128	123				
	136	143	125	130	122				
	140	135	123	122	135				
	139	124	117	135	137				
	137	129	123	138	130				
	134.2	133.2	122.8	129.7	126.7				

True dose rate (uR/h)	Average background counts(B) = 16.04						Corrected dose rate	Uncertainty	
	1	2	3	4	5	Avg			
118.95	121	117	139	120	119	119.9	103.86	12.68600252	
	118	125	132	106	120				
	120	128	124	105	118				
	113	122	128	109	119				
	112	120	130	110	122				
	118	119	126	117	119				
	117.0	121.8	129.8	111.2	119.5				

It is canbe seen that 2nd Orientation of the detector gives the closest reading to the true dose rate calculated above.

4.4 Response for Survey Meter

Cs-137		A = 3.7 MBq		Sept 2022		Distance = 50 cm					
True dose rate (uR/h)	Average background counts(B) = 0.016					Corrected dose rate	Uncertainty				
	1	2	3	4	5					Avg	
0.11895	0.09	0.14	0.1	0.15	0.16	0.12	0.104	12.56830601			
	0.14	0.11	0.12	0.11	0.1						
	0.10	0.14	0.11	0.08	0.09						
	0.14	0.10	0.12	0.09	0.12						
	0.14	0.09	0.09	0.15	0.16						
	0.09	0.15	0.1	0.09	0.14						
	0.12	0.12	0.11	0.11	0.13						
	0.12	0.12	0.11	0.11	0.13						

True dose rate (uR/h)	Average background counts(B) = 0.018					Corrected dose rate	Uncertainty		
	1	2	3	4	5				Avg
0.11895	0.12	0.16	0.09	0.15	0.14	0.12	0.102	14.24968474	
	0.11	0.14	0.12	0.12	0.15				
	0.17	0.12	0.09	0.09	0.14				
	0.10	0.09	0.17	0.11	0.1				
	0.14	0.15	0.14	0.14	0.14				
	0.09	0.10	0.11	0.1	0.09				
	0.12	0.13	0.12	0.12	0.13				
	0.12	0.13	0.12	0.12	0.13				

4.5 Response for Contamination Monitor

Cs-137	A = 3.7 MBq		Sept 2022		Distance = 50 cm					
True dose rate (uR/h)	Average background counts(B) = 16.04						Corrected dose rate	Uncertainty		
	1	2	3	4	5	Avg				
118.95	121.00	132.00	126.00	164.00	164.00	151.53	136.53	14.77091215		
	159.00	155.00	146.00	169.00	188.00					
	184.00	159.00	188.00	193.00	136.00					
	164.00	121.00	179.00	155.00	169.00					
	136.00	141.00	136.00	126.00	121.00					
	146.00	150.00	117.00	146.00	155.00					
	151.67	143.00	148.67	158.83	155.50					

4.6 True dose(D)

The true dose can be calculated using the formula:

$$D = R \cdot t \quad (9)$$

Where, R is the true dose rate in Sv/min and t is the time of exposure. Converting the true dose rate from $\mu\text{R/h}$ to Sv/min for 10 cm distance:

$$R = 26.13 \mu\text{Sv/hr} = 0.4355 \mu\text{Sv/min} \quad (10)$$

Now, calculating the true dose for 10 minutes of exposure:

$$D = 0.4355 \mu\text{Sv/min} \times 10 \text{ min} = 4.355 \mu\text{Sv} \quad (11)$$

$$D = 0.4355 \mu\text{Sv/min} \times 20 \text{ min} = 8.71 \mu\text{Sv} \quad (12)$$

4.7 Micro Digital Pocket Dosimeter

Cs-137	A = 3.7 MBq	Sept 2022	Distance = 10 cm		
Time (min)	True Dose (uSv)	Observed Dose(uSv)		Uncertainty (%)	
		MDD1 (798)	MDD2 (797)	MDD1	MDD2
10	4.35	4.00	4	8.12	8.12
20	8.71	8.00	9	8.12	-3.36



5 Conclusion

All tested instruments (Ion Chamber, GM Survey Meter, Contamination Monitor, and Pocket Dosimeters) provided readings within typical regulatory tolerance levels (generally $\pm 15\%$) relative to the calculated true dose rate. The Fluke 451P (in optimized orientation) and the Pocket Dosimeters demonstrated the highest degree of accuracy, validating their calibration for use in medical physics applications.

References